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# Productivity growth and real exchange rate dynamics: The Harrod-Balassa-Samuelson effects re-examined

*New empirical evidence on Harrod-Balassa-Samuelson effects in a combined sample of low-, middle-, and high-income countries using multivariate time series and panel data methods*

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**ABSTRACT**

This essay builds on the time-honoured Harrod-Balassa-Samuelson (HBS) effect and its role in explaining long run real exchange rate movements. We present an empirical analysis of the HBS effect in a modern context, as a productivity-bias approach to explaining deviations from PPP and in explaining dynamics of real exchange rates. Following Obstfeld and Rogoff (1996), we use a general equilibrium approach together with new data available from World Development Stats and IMF-IFS to investigate the empirical robustness of HBS effects using contemporary multivariate time series and panel data methods.

The HBS hypothesis is examined using a panel-data sample of 20 countries for the years 1991-2017. A simple general equilibrium model is constructed based on sectoral classifications and using assumptions of competitive markets, factor mobility and factor price equalisation. The analysis is conducted at three different levels- (a) country-specific analysis: Countries are analysed separately using time-series methods; (b) Panel data analysis: a combined panel of 20 countries is analysed; (c) separate panel analysis: panels of emerging and advanced economies are tested separately for presence of HBS effects. This study relies on currently prevalent time-series and panel data methods like Pesaran et al. (2001) ARDL bounds method and Pedroni (2004) for assessing cointegration between real exchange rates and the fundamentals. ARDL-OLS and panel-DOLS estimators are used for computing country-specific and panel estimates respectively. The analysis at each of the three different levels confirms the presence of a statistically significant long-run equilibrium relationship between real exchange rates and relative productivity in both advanced as well as emerging economies. The robustness of the HBS effect underlines the fact that even with growing financial interactions and interdependence among countries, the macroeconomic role of real factors like productivity nonetheless remains robust in a modern open economy and cannot be undermined.

**Keywords:** Harrod-Balassa Samuelson effect, real exchange rates, productivity, PPP, ARDL

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# 1 INTRODUCTION

The famous Purchasing Power Parity (PPP) theorem, first proposed in its modern form by Gustav Cassel, a Swedish economist, in 1921, was based on the law of one price, and stated that once converted to a common currency, price levels should be equal across countries in the long-run. In its purest version - the absolute PPP theorem suggests that the nominal exchange rates should adjust to offset any changes in a country's price level relative to that prevailing in other countries. In other words, the PPP theory holds that, in the long-run, relative price levels determine the exchange rate ($e = P/P^*$) and any deviation of relative prices from the equilibrium exchange rate will be ultimately 'mean-reverting'. PPP therefore implies that the real exchange rate (RER) should always equal 1 i.e. $P = eP^*$. The empirical failure of this relationship has continued to be a mainstream topic in international macroeconomics literature. Rogoff (1996) points out that "*While few empirically literate economists take PPP seriously as a short-term proposition, most instinctively believe in some variant of purchasing power parity as an anchor for long-run real exchange rates.*" Since, empirical studies have largely rejected the absolute PPP theorem, several attempts have been made to justify this failure. Reasons such as presence of trade and transaction costs, transportation costs, international goods market frictions, home-bias, inefficient and incomplete markets have been presented.

In this light, one of the most substantial explanations were given by Balassa (1964) and Samuelson (1964). Their well-known analyses explain the long-run behaviour of real exchange rates in terms of productivity differential between traded and non-traded goods in an economy. This was the first productivity-bias approach to explaining deviations from purchasing power parity and came to be known as the *Balassa-Samuelson effect.*[2] In simple words, this argument states that as productivity in traded goods sectors increases relative to that of non-traded goods sectors, the price levels in the non-traded goods sectors will rise, thereby causing aggregate price levels to rise and consequently generating a tendency for real exchange rate to appreciate.

The HBS effect not only explains the 'exchange rate effect' described above, but also the '*Penn effect'*. The Penn effect is the tendency for price levels to be higher in richer (higher GDP per capita) countries and that "*real per capita income ratios between poor and rich are systematically exaggerated by conventional exchange-rate conversions*." (Samuelson, 1994). While both these are two sides of the same coin, this essay explicitly examines the long-run relationship between traded-

[2] It is proposed that Roy Harrod (1933) was the first one to formulate this approach. Hence, some call it the *Harrod-Balassa-Samuelson effect*. Samuelson (1994) suggested a new name for this theory: the *Ricardo-Viner-Harrod-Balassa-Samuelson-Penn-Bhagwati effect.* The new convention among researchers is to use the name *Harrod-Balassa-Samuelson effect* (Tica and Družić, 2006) or in short, the ***HBS effect*** which is what is used throughout this essay.

sector relative productivity and real exchange rates in an empirical framework. This essay has a few salient features:

- The presence of this relationship is tested used newer data sets obtained from World Bank WDI database and IMF-IFS statistics. Evaluating this hypothesis using new data is essential since the past few decades have seen a surge in financial globalisation and increasing financial interaction and interdependence among countries. Hence, what was true in 1960s may not hold significant anymore. A question that we ask is whether real factors still play a significant role in modern open economies vis-à-vis nominal factors.
- Perhaps the most important objective of this essay is its attempt to (re-)establish the importance of real factors in exchange rate theories. The 21st century is accompanied by fast increasing financial integration across countries, rapidly growing cross-border financial flows and a boost to foreign direct and portfolio investments. Though both trade and financial flows might be necessary for explaining exchange rates, the explanatory power of nominal variables has evidently been strong. Consequently, monetary theories of exchange rates and new open economy macroeconomics theories have recently become popular in contemporary literature. This essay tries to establish that even though monetary factors may be crucial, real factors like productivity cannot be overlooked either.
- Instead of examining the HBS effect in a single country or in a set of advanced OECD coutries only, we investigate the HBS hypothesis in a panel-data sample consisting of a mix of 20 low-, middle- and high-income countries for the period 1991-2017. Literature on HBS effects in emerging economies is scant and handly any evidence supports presence of HBS effects in develing and emerging market economies.
- The analysis is conducted at three different levels-
  (a) Country specific analysis: 12 countries are analysed individually.
  (b) Panel-data analysis: 20 countries are analysed in a combined panel.
  (c) Finally, the countries are tested in subsets of advanced and emerging market economies.
- The study uses currently prevalent time-series and panel data econometric methods, namely, the ARDL bounds method and ARDL-OLS of Pesaran, Shin and Smith (2001), Pedroni (2004) and Kao and Chiang (2000) panel DOLS in obtaining country-specific and panel results respectively. Given that the HBS model became popular after 1964, a lot of empirical papers were published in the last quarter of the 20th century. However, most of such papers relied on conventional econometric methods like OLS, IV, LSDV, GMM, SUR, Engle-Granger cointegration test etc. in testing these models. Slowly these methods have been phased out and newer methods like FMOLS, DOLS, Johansen, ARDL, Pedroni etc. have become popular in econometric literature

since they deliver a higher level of consistency and efficiency. Application of ARDL bounds test and ARDL-OLS estimation in testing the empirical HBS model is a novel effort in this essay.[3]

- HBS theory has largely been ignored in modern theoretical as well as empirical literature. Choudhri and Khan (2004) point out that empirical testing of the HBS model has largely been restricted to developed countries and developing countries have been tested very rarely. To quote Tica and Družić (2006): *"...even today, the assertion of productivity biased purchasing power parity is considered an exotic and unconventional theory which is rarely used and almost never empirically confirmed. Textbooks on exchange rate economics usually only sporadically mention the theory and even contemporary promoters of the theory and researchers in the field maintain the idea that the HBS theory is poorly and rarely tested."* Thus, one of the objectives of this essay is to highlight the prominence of this theory and examining its empirical significance.

# 2 RELATED LITERATURE

There is an extensive literature providing theoretical underpinnings for the long-run determinants of RER, with the HBS model being a prominent one. Balassa (1964) and Samuelson (1964) independently argued this proposition. It was proposed that Ricardo (1911) and Harrod (1933) had already similarly argued before. In his famous study, Balassa (1964) regressed PPP based real exchange rates on per capita GNP for a cross-section of 12 industrial countries for the year 1960 using OLS and observed a positive and significant correlation between real exchange rates and p.c. GNP.

Rogoff (1992) was the first to formulate a mathematical HBS model using the general equilibrium framework. He uses the assumptions of perfect competition, perfect mobility of factors between sectors within the economy and the law of one price for traded goods; and shows that a change in the relative price in the non-traded sector does depend on a change in relative productivity of the sectors.

Gregorio, Giovannini and Wolf (1994) expand Rogoff's model by relaxing some extreme assumptions like competitive goods and factor markets and the law of one price for traded goods to bring in the demand side. They find inflation in non-traded sector to be higher than in traded sector and attribute this to faster productivity growth in the traded sector and a demand shift towards non-tradeables.

Gregorio and Wolf (1994) integrate terms of trade formally into the HBS model by using small open economy assumptions and use OECD country data to test the HBS proposition. They estimate the model using SUR in first differences and conclude "*the evidence broadly supports the predictions of*

[3] Tica and Družić (2006) do a detailed discussion on the evolution of the econometric HBS model.

*the model, namely that faster productivity growth in the tradable relative to the non-tradable sector and an improvement in the terms of trade induce a real appreciation.*"

Asea and Mendoza (1994) test the key propositions of the HBS model within a dynamic general equilibrium framework with balanced long-run growth path constraints. They show that along the BGP, the relative price of non-traded goods is determined by the ratio of the marginal products of labour in the traded and non-traded sectors. Their results indicate that labour productivity differentials strongly explain cross-country relative prices differences, however, non-traded relative prices are less successful in explaining deviations from PPP.

Alexius and Nilsson (2000) use the extended HBS model to establish that long-run real exchange rates are determined by relative productivity and terms of trade. They present evidence of this relationship in a dataset covering 15 OECD countries. They use Johansen and FMOLS techniques and their results are fairly in support of the model for panel data, though the support for time-series country level data is found to be weak.

Égert (2002) examines the HBS effects for a sample of east European countries during their transition process by using the Johansen test. He finds strong cointegration between productivity growth and relative prices, however the link between relative price movement and RER is found to be weaker.

Choudhri and Khan (2004) test the HBS model in a panel of 16 developing countries using Pedroni cointegration and DOLS estimation methods. They use a two-step estimation procedure by regressing relative prices on relative productivity and then regressing real exchange rates on relative prices. They decompose this relationship into two parts to make their model compatible with the dynamic new open economy macroeconomic models. Their findings strongly support the verification of HBS effects in developing countries.

Drine and Rault (2005) also use econometric methods like Pedroni and Johansen method to test the HBS model. While time series approach shows evidence against HBS hypothesis, non-stationary dynamic panel methods used by them show positive and significant results. They point out that a possible explanation of this part failure is that absolute PPP may not hold for traded goods as well.

Chong, Jordà and Taylor (2012) use panel-data methods to investigate the properties of the long-run PPP equilibrium using local projection methods for cointegrated systems. They find that short-run effects are absent, and the estimated movement towards the long-run equilibrium is much faster than traditionally thought. They conclude that due to the HBS effect the real exchange rate is converging not to a steady mean, but to a slowly moving target.

Hassan (2016) establishes evidence that the relationship between price levels and GDP is non-linear and turns negative among developing countries. According to him, the difference in stages of

development between countries can explain this finding. His finding indicates failure of HBS effect in developing countries and provides additional understanding of determinants of RER in these countries. A key feature of his paper is that he uses non-parametric estimation methods to arrive at this non-linear relationship between productivity and price levels.

Despite its simplicity and theoretical appeal, it is widely accepted in literature that the HBS model does not do well in explaining real exchange rates, except over very long-time horizons. A survey by Edwards and Savastano (2000) reviews a series of papers and concludes that "the Balassa-Samuelson effect is not borne out by the data". Froot and Rogoff (1994) give similar conclusions. A prominent survey study by Tica and Družić (2006) suggests that even though the HBS model has been put to a lot of empirical testing, the results are mixed and the support for the hypothesis is weak. They indicate that "*a growing body of empirical evidence points towards a need for rethinking about the significance of the HBS effect*." Officer (1976) completely dismisses the HBS proposition in his tests using cross-section data for 1950-1973. Arratibel, Rodriguez-Palenzuela and Thinman (2002) also reject the HBS hypothesis. Faria and León-Ledesma (2003) obtain empirical results that do not show supportive evidence for the HBS effect in the long-run. Fischer (2004) advocates alternative methods to fitting real exchange rate movements. Ricci, Milessi-Ferreti, and Lee (2013) also find weak evidence of HBS effects.

This essay follows a simple theoretical model closely resembling the one used by Gregorio and Wolf (1994). The model assumptions, data construction techniques and panel-data methods used here are similar to that of Choudhri and Khan (2004). However, this essay focuses on both advanced and emerging economies and establishes results at country-specific and panel-data levels, unlike Choudhri and Khan (2004) and most literature described above. It is pointed out that this essay has substantial findings in favour of the HBS model not only for advanced but for emerging countries as well, which is very rare in empirical HBS literature, both in terms of testing and favourable results.

Most empirical papers on the HBS model traditionally used econometric methods like OLS, SUR, IV, LSDV, RE etc. for estimation. These methods have been slowly phased out in newer empirical papers on HBS. This essay has tried to use some of the newer econometric methods like DOLS and ARDL, which have been proved to perform better in terms of consistency and efficiency in similar settings. The time-series estimation methods used in this essay, specifically, the ARDL method is relatively new in econometric literature and has not been applied to any of the empirical papers described above. Newer time-series empirical papers on the HBS model like Alexius and Nilsson (2000), Égert (2002), Drine and Rault (2005) have used the Johansen (1991) methods for non-stationary time-series data, however the Johansen method relies more on a large sample size and

requires $I(1)$ time-series. The ARDL method, works better in finite samples and does not necessarily rely on $I(1)$ time-series, and turns out to be empirically more useful for the HBS model.

Thus, this essay largely differs from existing empirical literature on the HBS model in terms of the theoretical model used, the assumptions, the econometric methods applied and the construction and use of datasets and variables. In terms of the results, this essay finds strong support of the presence of HBS effects in advanced as well as emerging countries, as opposed to some of the literature described above, which have rejected or found mixed results.

# 3 A SIMPLE HBS MODEL

This section outlines a simple theoretical framework for the HBS model following a general equilibrium approach as described in Obstfeld and Rogoff (1996). As long-run effects are the primary concern, short-run dynamics are nor modelled. The focus is on long-run equilibrium relations under complete wage-price adjustment. Consider a two country-two sector world with the two countries being home and foreign. Each country produces two goods- Tradable goods (*T*) and Non-Tradable good (*N*). Labour (*L*) and Capital (*K*) are used in the production of both the goods. We start with the following crucial assumptions:

- Labour markets are perfectly competitive. Labour $L = L_T + L_N$ is perfectly mobile across both sectors in each country but not internationally mobile. This implies that in equilibrium, wages must be equalalised in both sectors:

$$w_T = w_N = w$$

- Capital $K = K_T + K_N$ is perfectly mobile both domestically across sectors as well as internationally. This implies that $r_T = r_N = r = r^* = r^w$ where $r^w$ denotes the equilibrium world real interest rate.
- Cobb-Douglas Production functions for both goods:

$$Y_T = A_T K_T^{\alpha} L_T^{1-\alpha} \text{ and } Y_N = A_N K_N^{\alpha} L_N^{1-\alpha}, \text{ with } \alpha < 1.$$

  Another simplifying assumption that we make is that labour intensity $(1-\alpha)$ is equal in both the traded and non-traded sectors and the capital intensity $\alpha$ is also equal across sectors.

- Non-traded sector produces goods which are sold domestically only. The goods produced by the traded sector can be traded across countries. The law of one price holds for traded goods with $P_T = eP_T^*$ where $e$ denotes the home-foreign nominal exchange rate.
- Cobb-Douglas price level index is assumed: $P = P_T^{\gamma} P_N^{1-\gamma}$ and accordingly $P^* = P_T^{*\gamma} P_N^{*\,1-\gamma}$. The real exchange rate (RER) is then defined as $RER = \frac{P}{eP^*}$

- The nominal exchange rate is normalised: $e = 1$. The law of one price then simplifies to $P_T = P_T^*$. Making necessary substitutions in the *RER* relationship we get:

$$RER = \frac{P_T^{\gamma} P_N^{1-\gamma}}{P_T^{*\gamma} P_N^{*1-\gamma}} = \left(\frac{P_N}{P_N^*}\right)^{1-\gamma} \tag{1}$$

The Real exchange rate is thus obtained as a ratio of non-traded prices in home and foreign. The profit maximisation problem for firms in each sector $i$ where $i \in \{T, N\}$ is given as

$$\max_{K_i, L_i} \Pi_i = P_i Y_i - w_i L_i - r_i K_i$$

$$[K_T] \qquad \alpha P_T A_T \left(\frac{K_T}{L_T}\right)^{\alpha-1} = r_T \tag{2}$$

$$[L_T] \qquad (1-\alpha) P_T A_T \left(\frac{K_T}{L_T}\right)^{\alpha} = w_T \tag{3}$$

$$[K_N] \qquad \alpha P_N A_N \left(\frac{K_N}{L_N}\right)^{\alpha-1} = r_N \tag{4}$$

$$[L_N] \qquad (1-\alpha) P_N A_N \left(\frac{K_N}{L_N}\right)^{\alpha} = w_N \tag{5}$$

where $Y_i = A_i K_i^{\alpha} L_i^{1-\alpha}$ and FOCs are as follows:

Combining the FOCs with factor price equalisation across sectors assumptions:

$$w_T = w_N = w \tag{6}$$

$$r_T = r_N = r = r^* = r^w \tag{7}$$

Using all these results and making necessary substitutions yields:

$$P_T A_T = P_N A_N \text{ or equivalently, } \frac{P_N}{P_T} = \frac{A_T}{A_N} \tag{8}$$

$$w = (1-\alpha)\left(\frac{\alpha}{r^w}\right)^{\frac{\alpha}{1-\alpha}} (P_T A_T)^{\frac{1}{1-\alpha}} \tag{9}$$

Eq. 9 shows a strongly linked positive relationship between traded sector productivity $A_T$ and the equilibrium wages *w*, given the world real interest rate $r^w$. Eq. 8 shows that an increase in relative productivity in the traded sector must evoke an increase in relative prices in the non-traded sector as wages will rise in the non-traded sector due to the relationship in eq. (9). In other words, an increase in $A_T$ causes wages in the traded sector to increase. Perfect mobility of labour across sectors implies that this increase in traded-sector wages will push workers out from the non-traded sector. The only way for the non-traded sector to continue existing is to offer the same higher wages to its workers which the traded sector is offering. Thus, the wages in the non-traded sector also increase until wages in both sectors are equalised at a new higher level. Since $A_N$ has not changed and eq. (5) must hold

in equilibrium, an increase in wages in the non-traded sector must be accompanied by an increase in non-traded sector prices $P_N$, *ceteris paribus*. This is the key proposition of the Harrod-Balassa-Samuelson hypothesis.

A first implication is the *Penn effect* i.e. richer countries with higher productivity in the traded sectors also tend to have higher prices in the non-traded sectors and hence higher overall price levels. The classical textbook example offered is that a haircut (which is a classic example of a non-traded good) costs much more in a rich country like UK than in a poorer country like India. The Penn effect is easily observable when travelling internationally.

A second implication of the HBS hypothesis is that as traded sector productivity growth causes non-traded sector prices to rise (and hence overall price levels to rise), the real exchange rate of the country also appreciates. To see this, we combine eq. (1) with eq. (8) and the corresponding eq. (8) for foreign.

$$\frac{P_N}{P_T} = \frac{A_T}{A_N} \qquad (8) \qquad\qquad \frac{P_N^*}{P_T^*} = \frac{A_T^*}{A_N^*} \qquad (8^*)$$

$$\frac{P_N}{P_N^*} = \frac{A_T}{A_T^*} \cdot \frac{A_N^*}{A_N} \qquad (10)$$

Dividing (8) by (8*) and making use of the law of one price in the traded sector i.e. $P_T = P_T^*$:

Combining eq. (10) with (1) yields:

$$RER = \left(\frac{A_T}{A_T^*} \cdot \frac{A_N^*}{A_N}\right)^{1-\gamma} \qquad (11)$$

This relationship shows that an increase in traded sector productivity in home, $A_T$, *ceteris paribus,* causes home's real exchange rate to appreciate. Next, we take natural logarithms in eq. (11) both sides, which gives us the deterministic empirical model

$$\ln RER = (1-\gamma).\left[\ln\left(\frac{A_T}{A_T^*}\right) - \ln\left(\frac{A_N}{A_N^*}\right)\right]$$

Another key simplifying assumption to make at this stage is that productivity in the non-traded sector in both home and foreign are same i.e. $A_N = A_N^*$. This reduces the above equation to:

$$ln\, RER = (1-\gamma).\, ln\left(\frac{A_T}{A_T^*}\right) \qquad (12)$$

Theoretically, $(1-\gamma)$ is expected to be positive. A positive and statistically significant estimate of $(1-\gamma)$ would be empirical proof in favour of the Harrod-Balassa-Samuelson hypothesis i.e. an increase in relative productivity in the traded sector induces real appreciation in home.

Eq. (12) is re-written below in a stochastic form to make it empirically estimable.

$$ln\, RER_t = \beta_0 + \beta_1 . ln\left(\frac{A_{T,t}}{A^*_{T,t}}\right) + \varepsilon_t \qquad (13)$$

We follow the approach similar to De Gregorio and Wolf (1994) and add terms of trade as the second explanatory variable in the HBS model as a control for home bias and absence of complete PPP in traded sectors. While it has been omitted in our theoretical model for simplicity, we add it in the empirical model. Choudhri and Khan (2004) explicitly theoretically model terms of trade. We directly use an empirical equation similar to the one used by De Gregorio and Wolf (1994) and Choudhri and Khan (2004).

$$ln\, RER_t = \beta_0 + \beta_1 . ln\left(\frac{A_{T,t}}{A^*_{T,t}}\right) + \beta_2 . ln\, TOT_t + \varepsilon_t \qquad (14)$$

Before proceeding further, two limiting assumptions in this theoretical model need to be justified further. First, is the assumption that factor income shares are equal in the traded and on-traded sectors. Obstfeld and Rogoff (1996) explicitly build a theoretical model by assuming separate factor income shares in the traded and non-traded sectors. This does not change the overall analysis much but brings in an additional coefficient in eq. (12).

A second assumption that needs discussion is that productivity in the non-traded sector in both home and foreign are same i.e. $A_N = A^*_N$. Although it seems like a limiting assumption it has been quite popular in empirical HBS model papers. Moreover, a classic example that makes this assumption plausible is the realistic observation that non-traded sector technology like haircutting technology is almost same in all types of countries, however, prices vary considerably. Problems with sectoral data availability across countries and classification of sectors in tradable and non-tradable categories are also major issues here. In fact, this assumption implies that relative to traded sector, non-traded sector productivity growth rate can be negligible.

# 4 DATA AND VARIABLE DESCRIPTION

The data set consists of 20 countries with a good mix of low-, middle- and high-income economies.[4] The countries are selected based on (i) ease of data availability; (ii) to ensure representation of countries from across all income groups; (iii) most of the countries in the panel belong to either OECD, G20, BRICS or other major economic organisations. The sample period is 1991-2017 at annual frequency. The complete panel therefore consists of 20 cross-sectional units each with 27 sample points.

IMF's Consumer Price Index (CPI) based Real Effective Exchange Rate (REER) is used as the dependent variable. The data for REER is collected from World Bank's WDI dataset.[5] The data for the independent variable terms of trade (TOT) is also sourced from World Bank's WDI database from the series 'Net Barter Terms of Trade'.[6]

The right choice of a proxy for productivity in testing the HBS hypothesis (and in general) has been a source of great controversy in literature. GDP, GDP-per-capita, GDP-per-capita-per-hour, labour productivity and total factor productivity (TFP) are commonly used ones. Though average labour productivity is the far most popular one, many have argued in favour of TFP being a relatively better measure. However, sectoral TFP data is almost non-existent and times-series for countries vary considerably in length due to data measurement, depth and consistency issues. This essay follows Choudhri and Khan (2004) and uses labour productivity in agriculture and industry sectors as a proxy for traded sector productivity $A_T$. World average labour productivity in agriculture and industry is used as a proxy for $A_T^*$. This is in line with the assumption that bulk of goods in the industry and agriculture sectors are traded, while the services sector produces bulk of non-traded goods in any country.[7] Labour productivity in agriculture and industry is obtained from World Bank WDI database from the series 'Value added per worker in agriculture and industry'.[8]

The variables used in regression are described as follows:

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[4] This is as per World Bank classification, June 2018. According to the World Bank: "*Low-income economies are defined as those with a GNI per capita, of $995 or less in 2017; lower middle-income economies are those with a GNI per capita between $996 and $3,895; upper middle-income economies are those with a GNI per capita between $3,896 and $12,055; high-income economies are those with a GNI per capita of $12,056 or more.*"

In our sample: high income countries: Argentina, Australia, Chile, France, Germany, Norway, Singapore, South Korea, United Kingdom; upper-middle income countries: Brazil, Mexico, South Africa; lower-middle and low-income countries: Bangladesh, Ethiopia, India, Indonesia, Kenya, Nigeria, Pakistan and Vietnam.

[5] See Appendix 1 for details on IMF's CPI based real effective exchange rate (REER).

[6] See Appendix 1 for TOT index calculation.

[7] Tica and Družić (2006) discuss more on this.

[8] See Appendix 1 for detailed calculation of relative traded sector productivity proxy.

- Natural logarithm of real effective exchange rate is denoted as LREER.
- Natural logarithm of traded sector productivity index is denoted as LRPRO.
- Natural logarithm of terms of trade index is denoted as LTOT.[9]

For country specific analysis, the empirical model is:

$$LREER_t = \beta_0 + \beta_1 . LRPRO_t + \beta_2 . LTOT_t + \varepsilon_t \qquad (15)$$

For panel-data analysis, the empirical model is:

$$LREER_{it} = \alpha_i + \delta_t + \beta_1 . LRPRO_{it} + \beta_2 . LTOT_{it} + u_{it} \qquad (16)$$

$\beta_1$ denotes the Harrod-Balassa-Samuelson effect, it is the key parameter of interest and is theoretically expected to be positive. $\beta_2$ denotes the terms of trade effect. As the models are in log-linear form, $\beta_1$ and $\beta_2$ can be interpreted as long-run elasticity coefficients. Simply, a 1% increase in the relative productivity index should associate with $\beta_1$% change in the equilibrium real exchange rate in the long-run.

# 5 ECONOMETRIC METHODOLOGY

## 5.1 COUNTRY SPECIFIC ANALYSIS

The country specific analysis is conducted on data from 12 different countries. The first step in estimating eq. (15) is to check whether LREER, LRPRO and LTOT are stationary, to avoid the possibility of spurious regressions. Therefore, for each country unit root test is performed on each of the three time-series LREER, LRPRO and LTOT. Augmented Dickey-Fuller (ADF) unit root test method is used.[10] Tests are performed on the levels and first differences of these series. The purpose of the unit root tests is to ensure that the time-series are either $I(1)$ or $I(0)$ but not $I(2)$ because the next step in the process is to test for cointegration among the series using the ARDL bounds test and a necessary condition for this is that none of the series should be $I(2)$.

If the desired results are obtained from unit root tests, we proceed to test for the presence of a long-run cointegrating relationship between LREER, LRPRO and LTOT for each country. The Autoregressive Distributed Lag (ARDL) bounds test based on Pesaran, Shin and Smith (2001) is used

[9] See Appendix 1 for calculation of LREER, LRPRO, LTOT.
[10] See Appendix 2 for details of ADF unit root test.

for this purpose.[11] [12] The ARDL bounds test is advantageous since it is useable if the variables are either $I(0)$ or $I(1)$ $\left(\text{but not } I(2)\right)$. It is also simpler in the sense that it is based on OLS. Thirdly, it is relatively more efficient in small or finite sample data sizes as is the case in this study. The AEG test, a traditional cointegration test, is also employed partially in this essay if the ARDL bounds test for a given country is inconclusive or does not reject the null.[13]

In case the ARDL bounds test confirms cointegration, the long-run cointegrating vector (in other words, the estimates of $\beta_1$ and $\beta_2$ in eq. (15)) is obtained from the OLS estimation of a conditional error correction model (CECM). The estimators of the long-run parameters are super-consistent and asymptotically normally distributed irrespective of the order of integration of the regressors.[14]

In case, the ARDL test fails in a given country but the AEG test confirms cointegration, the cointegrating vector is estimated using Dynamic Ordinary Least Squares, which according to Stock and Watson (1993), is more efficient than OLS, when used with HAC standard errors and the t-statistics have a standard normal distribution asymptotically.[15]

## 5.2 PANEL ANALYSIS

Similar to the country specific analysis, panel data estimation using OLS could be a spurious regression if the panel series LREER, LRPRO and LTOT are non-stationary. Therefore, the series are tested for presence of unit roots using two popular panel data unit root tests- Im-Pesaran-Shin test (IPS, 2003) and Breitung test (2000).[16]

If the unit root tests indicate non-stationarity in levels i.e. if LREER, LRPRO and LTOT are found to be $I(1)$, we proceed to panel cointegration testing to detect if there is a long-run equilibrium relationship among the series. Kao (1999) and Pedroni (2004) panel cointegration tests are performed.[17] If cointegration is confirmed, the final step is to estimate the homogeneous long-run

[11] See Appendix 2 for details of PSS (2001) ARDL bounds test.

[12] In contemporary time-series econometrics a few cointegration tests are quite popular. The most popular among these is the Johansen cointegration test based on Johansen (1991) and Augmented Engle-Granger (AEG) Test. While the ARDL bounds test and the AEG tests are single equation cointegration tests, the Johansen test is a multivariate cointegration test and uses a vector error correction model (VECM) to test for the presence of multiple cointegrating relationships simultaneously. We would have preferred to use the Johansen test in our analysis, however: (a) this test is a large sample test, whereas the sample size (for each country) in our analysis is very small i.e. only 27; (b) a necessary condition for this test is that all time series should be $I(1)$ which may not be the case in our analysis.

[13] See Appendix 2 for details of two step AEG test.

[14] See Appendix 2 for details of the ARDL bounds test and ARDL-OLS estimation procedure.

[15] See Appendix 2 for details of Stock and Watson's DOLS estimaton.

[16] See Appendix 2 for details of IPS test and Breitung test.

[17] See Appendix 2 for details of Kao and Pedroni tests.

cointegrating vector for the panel series. Estimation is done using the panel Dynamic Ordinary Least Squares (DOLS).[18]

# 6 RESULTS

## 6.1 COUNTRY SPECIFIC RESULTS

### 6.1.1 ADF UNIT ROOT TESTS

For each country $i$ ADF Unit unit root test results on LREER, LRPRO and LTOT are presented in Table 1. The test is performed on the levels and on the first differences for each series.

The LREER series: (a) In India is stationary ($I(0)$) as the ADF test statistic is significant at 1%. (b) In South Africa, the statistic is significant at 10%. For all other countries, the null hypothesis of a unit root in LREER series in level cannot be rejected.

Similarly, the LRPRO series: (a) In Vietnam is stationary ($I(0)$) as the ADF test statistic is significant at 1%. (b)In Australia, the statistic is significant at 10%. For all other countries, the null hypothesis of a unit root in LRPRO series in level cannot be rejected.

Finally, the LTOT series: (a) In Germany can be considered stationary ($I(0)$) as the ADF test statistic is significant at 5%. (b)In Vietnam and Indonesia, the statistic is significant at 10%. For all other countries, the null hypothesis of a unit root in LTOT series in level cannot be rejected.

The first differences of LREER, LRPRO and LTOT for all countries reject the null of a unit root as the tau-statistics are significant at 5% (in fact at 1% for most). Thus, the first differences appear to be stationary.

We conclude that the series LREER, LRPRO and LTOT are $I(1)$ for most countries, except for the cases mentioned above. In other words, LREER, LRPRO and LTOT are mostly $I(1)$ but $I(0)$ in just a few cases. Since none of the series are $I(2)$ we can proceed with the ARDL bounds cointegration tests.

[18] It is pointed out that there is not much consensus in literature for estimation of non-stationary panel cointegrated series. Kao and Chiang (2000) have shown that OLS is inconsistent in panel data, unlike the time-series counterpart. Alternatives like FMOLS, ARDL and DOLS have been proposed. The DOLS procedure is used in this essay. DOLS explicitly adds leads and lags of the first differences of the regressors to the model. Kao and Chiang (2000) and Pedroni (2001) have extended DOLS to panel setting and shown it is an improvement over FMOLS. Since first-difference terms are included in this procedure, the coefficients of level terms represent long-run effects and the coefficients on the differenced terms represent the short run dynamics. The procedure allows for heterogeneous short run dynamics deterministics (constant or trend) across cross-sections. See Appendix 2 for details of Panel DOLS estimation.

**Table 1: ADF Unit Root Test Results**

| COUNTRY | | LREER | | LRPRO | | LTOT | |
|---|---|---|---|---|---|---|---|
| | | Level | First Diff. | Level | First Diff. | Level | First Diff. |
| AUSTRALIA | τ-statistic | -2.53298 | -3.922877*** | -2.69798* | -6.16911*** | -1.833319 | -4.84508*** |
| | Prob. | 0.3110 | 0.0063 | 0.0879 | 0.0000 | 0.3569 | 0.0007 |
| BANGLADESH | τ-statistic | 0.817925 | -3.230197*** | -1.960895 | -5.603442*** | -1.061073 | -5.033751*** |
| | Prob. | 0.9924 | 0.0024 | 0.3011 | 0.0001 | 0.7153 | 0.0004 |
| BRAZIL | τ-statistic | -1.765468 | -4.391539*** | -2.322232 | -5.859546*** | -3.130006 | -3.486599** |
| | Prob. | 0.3883 | 0.0021 | 0.1729 | 0.0001 | 0.1213 | 0.0171 |
| ETHIOPIA | τ-statistic | -1.408155 | -4.110115*** | -1.436423 | -4.568344* | -1.756591 | -2.565481** |
| | Prob. | 0.8343 | 0.0041 | 0.5483 | 0.0014 | 0.3912 | 0.0129 |
| GERMANY | τ-statistic | -2.634592 | -4.210399*** | -2.137555 | -5.043425*** | -3.274158** | -4.805457*** |
| | Prob. | 0.2693 | 0.0032 | 0.2325 | 0.0004 | 0.0268 | 0.0008 |
| INDIA | τ-statistic | -5.225935*** | -7.036921*** | -1.534628 | -5.664438*** | -1.847134 | -4.472984*** |
| | Prob. | 0.0014 | 0.0000 | 0.7907 | 0.0001 | 0.3506 | 0.0017 |
| INDONESIA | τ-statistic | -2.332525 | -5.22032*** | -2.14729 | -5.210254*** | -3.414195* | -5.321987*** |
| | Prob. | 0.1702 | 0.0003 | 0.2291 | 0.0003 | 0.0712 | 0.0002 |
| NORWAY | τ-statistic | -1.523651 | -4.186805*** | -2.596224 | -4.664128*** | -1.373634 | -4.423336*** |
| | Prob. | 0.5061 | 0.0034 | 0.2847 | 0.0011 | 0.8448 | 0.0019 |
| SINGAPORE | τ-statistic | -1.665514 | -2.908048*** | -1.969329 | -5.586927*** | -2.071534 | -3.881732*** |
| | Prob. | 0.4357 | 0.0054 | 0.2976 | 0.0001 | 0.2569 | 0.0069 |
| SOUTH AFRICA | τ-statistic | -3.363859* | -4.410879*** | -2.451914 | -4.733852*** | -1.997502 | -3.35717** |
| | Prob. | 0.0793 | 0.0020 | 0.1383 | 0.0009 | 0.5744 | 0.0228 |
| SOUTH KOREA | τ-statistic | -2.233778 | -3.849605*** | -1.689519 | -5.352352*** | -0.970937 | -3.554754** |
| | Prob. | 0.2001 | 0.0075 | 0.7269 | 0.0002 | 0.7482 | 0.0147 |
| VIETNAM | τ-statistic | -1.935307 | -4.629156*** | -5.425447*** | -7.956882*** | -3.426887* | -3.923819*** |
| | Prob. | 0.6077 | 0.0012 | 0.0009 | 0.0000 | 0.0760 | 0.0063 |

**Notes: (1)** The Mackinnon (1996) p-values are shown. **(2)** *** denotes significant at 1%; ** denotes significant at 5%; * denotes significant at 10%. **(3)** ADF test has three different model specifications for deterministics- (a) no drift, no trend; (b) only drift (constant); (c) drift and trend; Either (b) or (c) is used in unit root testing in the levels of all series. Accordingly, specification (a) or (b) is used for unit root testing in first differences of all series based on statistical significance of the deterministics. **(4)** Automatic lag length selection based on Schwarz Information Criteria (SIC) is used with automatic selection of maximum lags.

### 6.1.2 COINTEGRATION TESTS

Table 2 presents the ARDL bounds cointegration test results (based on Pesaran *et al*, 2001) for all countries. If both the test statistics- F-statistic and the t-statistic are significant at 10% or 5%, the null hypothesis of no cointegration is rejected. The results show that except for 3 countries, the null is rejected in all other countries and a long-run cointegrating relationship between LREER, LRPRO and LTOT can be concluded. In Ethiopia, the F-statistic is inconclusive at 10% and the t-statistic cannot reject the null. Hence, we conclude no cointegration in Ethiopia using this test. In India, the F-statistic is significant at 10% but the t-statistic is inconclusive. In South Africa, the F-statistic is inconclusive at 10% but the t-statistic is significant. Therefore, the test remains inconclusive for both South Africa and India.

**Table 2: ARDL Bounds Cointegration Test**

| COUNTRY | Optimal Lags (SIC) | Model[19] | Bounds Test Statistic | | Prob. | | CONCLUSION |
|---|---|---|---|---|---|---|---|
| | | | | | $I(0)$ | $I(1)$ | |
| **AUSTRALIA** | 2,1,0 | 5 | F-stat | 7.788546** | 0.015 | 0.035 | COINTEGRATION |
| | | | t-stat | -4.751694** | 0.005 | 0.018 | |
| **BANGLADESH** | 2,2,0 | 5 | F-stat | 7.171827* | 0.022 | 0.051 | COINTEGRATION |
| | | | t-stat | -3.890462* | 0.027 | 0.075 | |
| **BRAZIL** | 4,0,1 | 3 | F-stat | 5.723078* | 0.023 | 0.060 | COINTEGRATION |
| | | | t-stat | -3.809462** | 0.011 | 0.042 | |
| **ETHIOPIA** | 4,0,0 | 4 | F-stat | -3.049638 | 0.053 | 0.109 | NO COINTEGRATION |
| | | | t-stat | 4.785371 | 0.117 | 0.235 | |
| **GERMANY** | 1,2,4 | 5 | F-stat | 8.947808** | 0.013 | 0.031 | COINTEGRATION |
| | | | t-stat | -4.646953** | 0.009 | 0.027 | |
| **INDIA** | 1,3,1 | 5 | F-stat | 7.110308* | 0.025 | 0.057 | INCONCLUSIVE |
| | | | t-stat | -3.692437 | 0.039 | 0.101 | |
| **INDONESIA** | 1,1,0 | 5 | F-stat | 11.34223*** | 0.002 | 0.005 | COINTEGRATION |
| | | | t-stat | -5.744598*** | 0.001 | 0.003 | |
| **NORWAY** | 2,0,3 | 1 | F-stat | 3.993729* | 0.025 | 0.078 | COINTEGRATION |
| | | | t-stat | -3.076296* | 0.006 | 0.055 | |
| **SINGAPORE** | 3,4,1 | 3 | F-stat | 5.811444* | 0.024 | 0.064 | COINTEGRATION |
| | | | t-stat | -3.962857** | 0.009 | 0.035 | |
| **SOUTH AFRICA** | 2,1,0 | 3 | F-stat | 4.570156 | 0.044 | 0.106 | INCONCLUSIVE |
| | | | t-stat | -3.615911* | 0.013 | 0.055 | |
| **SOUTH KOREA** | 2,0,2 | 1 | F-stat | 4.795385** | 0.012 | 0.041 | COINTEGRATION |
| | | | t-stat | -3.479635** | 0.002 | 0.026 | |
| **VIETNAM** | 1,0,3 | 2 | F-stat | 4.85644** | 0.019 | 0.050 | COINTEGRATION |
| | | | t-stat | -3.418962* | 0.021 | 0.076 | |

**Notes: (1)** *** denotes significant at 1%; ** denotes significant at 5%; * denotes significant at 10%. **(2)** The optimal lags for each country are selected automatically using SIC. The maximum lag varies from 1 to 4 across countries. **(3)** The Bounds test has 5 different model specifications for deterministics. The model column shows the selected model for each country. The details for this are given in Appendix 2. **(4)** Approximate p-value bounds based on Kripfganz and Schneider (2018) critical values.

### 6.1.3 ESTIMATION RESULTS

In this section we present the long-run estimates of the parameters $\beta_1$ *and* $\beta_2$ from eq. (15) for each country. These estimates are obtained from the ARDL OLS estimation procedure with the model and lag specifications for each country as described in Table 2. Table 3 presents the estimates of the long-

[19] Details of ARDL model deterministics specifications in Appendix 2.

run parameters obtained from the ARDL OLS estimation procedure for each of the 9 countries where the bounds test was conclusive.

The coefficient on LRPRO ($\beta_1$) for each country represents the Harrod-Balassa-Samuelson effect as described earlier. From Table 3 we can see that the coefficients are positive as expected and statistically significant at 1% or 5%, without any exceptions. The coefficients can be interpreted as long-run elasticity coefficients. For e.g., in Australia, the results show that a (permanent) increase in relative productivity by 1% explains a 0.49% increase (appreciation) in Australia's real exchange rate in the long-run. We have similar interpretations for other countries. From these results we can conclude strongly in favour of the HBS effect.

**Table 3: ARDL OLS Estimation results**

| COUNTRY | Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|---|
| AUSTRALIA | LRPRO | 0.494159** | 0.185787 | 2.659814 | 0.0160 |
| | LTOT | 0.209567*** | 0.05268 | 3.978084 | 0.0009 |
| BANGLADESH | LRPRO | 0.837681*** | 0.262173 | 3.195147 | 0.0053 |
| | LTOT | 0.682141*** | 0.109433 | 6.233444 | 0.0000 |
| BRAZIL | LRPRO | 0.522444** | 0.181555 | 2.877607 | 0.0115 |
| | LTOT | 0.913235*** | 0.280221 | 3.258978 | 0.0053 |
| GERMANY | LRPRO | 0.818357*** | 0.148207 | 5.521721 | 0.0001 |
| | LTOT | 1.698953* | 0.869154 | 1.954721 | 0.0743 |
| INDONESIA | LRPRO | 0.707233*** | 0.187226 | 3.777436 | 0.0012 |
| | LTOT | 0.618516*** | 0.2128 | 2.906556 | 0.0087 |
| NORWAY | LRPRO | 0.519876*** | 0.024401 | 21.30541 | 0.0000 |
| | LTOT | 0.163386*** | 0.03804 | 4.295079 | 0.0005 |
| SINGAPORE | LRPRO | 0.48980*** | 0.12246 | 3.999682 | 0.0018 |
| | LTOT | -0.43997* | 0.231954 | -1.8968 | 0.0822 |
| SOUTH KOREA | LRPRO | 0.550351*** | 0.066331 | 8.297088 | 0.0000 |
| | LTOT | 0.44115*** | 0.0803 | 5.493775 | 0.0000 |
| VIETNAM | LRPRO | 1.046789** | 0.391611 | 2.673037 | 0.0161 |
| | LTOT | 1.658241*** | 0.281075 | 5.899648 | 0.0000 |

**Notes: (1)** Method: ARDL OLS estimation**. (2)** *** denotes significant at 1%; ** denotes significant at 5%; * denotes significant at 10%.

The coefficient on LTOT ($\beta_2$) for each country represents the terms of trade effect. The coefficients are positive and statistically significant for all countries, except Singapore, where the coefficient is negative but significant at 10%.

In Ethiopia, India and South Africa, the ARDL bounds test was either inconclusive or could not reject the null of no cointegration. We make a second attempt at testing cointegration in these countries

using the traditionally used two-step Augmented Engle Granger Test. Table 4 gives the AEG cointegration test results.

**Table 4: AEG Test**

| COUNTRY | tau-statistic | Prob. | CE Spec.† |
|---|---|---|---|
| ETHIOPIA | -2.633853 | 0.2312 | a |
| INDIA | -5.924994*** | 0.0039 | c |
| SOUTH AFRICA | -4.447013* | 0.0674 | c |

**Notes: (1)** *** denotes significant at 1%; ** denotes significant at 5%; * denotes significant at 10%. **(2)** Automatic lag selection based on SIC. **(3)** † The AEG test has 4 different model specifications for deterministics in the cointegrating equation. (a) no constant, no trend; (b) constant; (c) linear trend; and (d) quadratic trend. The column above lists the selected specification. **(4)** MacKinnon (1996) p-values.

The AEG test shows that for South Africa and India we can reject the null hypothesis of no cointegration at 10% and 5% respectively. The null cannot be rejected for Ethiopia. Thus, both bounds and AEG tests suggest no cointegration in Ethiopia. This could be due to a model misspecification or due to a small sample size. Since cointegration is confirmed in India and South Africa, we obtain the long-run cointegrating coefficients using DOLS. The results are shown in table 5.

**Table 5: DOLS estimation**

| COUNTRY | Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|---|
| **INDIA** | LRPRO | -0.139744 | 0.093774 | -1.490218 | 0.1518 |
| | LTOT | 0.284315*** | 0.055929 | 5.083540 | 0.0001 |
| **SOUTH AFRICA** | LRPRO | 0.764791*** | 0.106832 | 7.158833 | 0.0000 |
| | LTOT | -0.556261*** | 0.180483 | -3.082076 | 0.0064 |

**Notes: (1)** Method: DOLS **(2)** *** denotes significant at 1%; ** denotes significant at 5%; **(3)** HAC (Newey-West) standard errors **(4)** Automatic leads and lags specification using SIC with max = 1.

The coefficient of LRPRO for South Africa is statistically significant and with expected sign, showing empirical proof in favour of the HBS effect. However, the TOT effect is negative but significant. For India, though the TOT effect is positive and significant, the HBS coefficient is negative and insignificant. Thus, for India, the data cannot justify presence of HBS effects. However, this result needs to be interpreted cautiously due to small data sample or a possible model misspecification.

To conclude the country-specific analysis, we can summarise that out of 12 countries tested for the presence of the Harrod-Balassa-Samuelson effects, data support its existence in 10 of these countries. In Ethiopia and India, HBS effects could be verified in data. One must note that both these are emerging market counties. There is strong evidence in favour of HBS effects in other countries, irrespective of their income/growth classification. This result is consistent with some of the previous studies described in section 2.

## 6.2 PANEL RESULTS

### 6.2.1 UNIT ROOT TESTS

The combined panel data analysis is conducted on a balanced panel of 20 countries for the years 1991-2017. This section presents the results of the IPS unit root test and the Breitung test to check if the LREER, LRPRO and LTOT contain unit roots. The tests are performed on the levels and the first differences of the series. Table 6 shows the results.

The unit root tests strongly suggest that each of the series are $I(1)$. The unit root test on the first differences of the series strongly reject the null. The test on the levels cannot reject the null of a unit root for all the series. We conclude that the levels of all the series are non-stationary, but the first differences are stationary. Since all the series appear to be $I(1)$, we can proceed to test for cointegration among them.

**Table 6: Panel Unit Root Tests**

| Panel Unit Root Test | LREER | | LRPRO | | LTOT | |
|---|---|---|---|---|---|---|
| | Level | First Diff. | Level | First Diff. | Level | First Diff. |
| **Im, Pesaran and Shin W-stat** | -0.3141 | -14.8937*** | 0.24647 | -18.3993*** | -0.46276 | -14.1144*** |
| **Prob.** | 0.3767 | 0.0000 | 0.5973 | 0.0000 | 0.3218 | 0.0000 |
| **Breitung t-stat** | -0.7064 | -10.0939*** | 0.16303 | -13.1813*** | -0.63234 | -10.4532*** |
| **Prob.** | 0.2400 | 0.0000 | 0.5648 | 0.0000 | 0.2636 | 0.0000 |

**Notes: (1)** *** denotes significant at 1%; **(2)** Automatic lags selection based on SIC; Automatic selection of maximum lags. **(3)** Exogenous variables specification in levels: Individual effects, individual linear trends **(4)** Exogenous variables specification in first diff: Individual effects.

### 6.2.2 COINTEGRATION TESTS

The unit root tests on LREER, LRPRO and LTOT suggest that they are integrated of order 1. This section presents the results of the panel cointegration tests on these series. As described above, Kao test (1999) and Pedroni test (2004) are used for testing cointegration. The table below shows the Kao ADF type t-statistic. Out of the seven test statistics in Pedroni test, the parametric ADF statistic and the non-parametric PP statistic are reported for both the within dimension (panel test) and the between dimension cases (group test) are reported.

The results show strong support for a long-run cointegrating relationship between LREER, LRPRO and LTOT. Kao t-statistic is significant at 5% and rejects the null. The panel PP and ADF statistics of the Pedroni test are strongly significant at 1% and reject the null in favour of a homogeneous AR coefficient for all cross-sections. The group PP and ADF statistics are also strongly significant at 1% and reject the null in favour of heterogeneous AR coefficients for the cross-sections. For our analysis,

the between dimension test is more relevant since it is less restrictive. Nonetheless, all tests strongly support cointegration. Since the non-stationary series LREER, LRPRO and LTOT appear to be panel-cointegrated.

**Table 7: Panel Cointegration Tests**

| Cointegration Test | Statistic | Prob. | |
|---|---|---|---|
| **Kao** | t-Statistic | -2.220156** | 0.0132 |
| **Pedroni (within dimension)** | Panel PP-Statistic | -2.882833*** | 0.0020 |
| | Panel ADF-Statistic | -3.046199*** | 0.0012 |
| **Pedroni (between dimension)** | Group PP-Statistic | -2.482916*** | 0.0065 |
| | Group ADF-Statistic | -3.831248*** | 0.0001 |

**Notes: (1)** *** denotes significant at 1%; ** denotes significant at 5%; **(2)** Automatic lags selection based on SIC; Automatic selection of maximum lags. **(3)** Pedroni test determisitics: intercept and trend. **(4)** Kao test only allows for intercept as the deterministic.

### 6.2.3 Estimation Results

Table 8 presents the estimation results of the HBS model i.e. eq. (16) for a panel sample of 20 countries, using the panel grouped DOLS estimator. HAC standard errors used.

**Table 8: Panel Estimation Results**

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| LRPRO | 0.357923*** | 0.091801 | 3.898922 | 0.0001 |
| LTOT | 0.699602*** | 0.139722 | 5.007106 | 0.0000 |

**Notes: (1)** Dependent Variable: LREER **(2)** Method: Grouped Panel Dynamic Least Squares (DOLS) **(3)** *** denotes significant at 1%; **(4)** Cointegrating equation deterministics: constant, linear trend; **(5)** Fixed leads and lags specification (lead=1, lag=1) **(6)** Individual HAC standard errors & covariances.

The estimated coefficient on LRPRO $(\beta_1)$ is positive and statistically significant. As described above, it can be interpreted as a long-run elasticity coefficient and shows the average effect of (permanent) relative productivity growth on long-run real exchange rates. In other words, we can say, an increase in relative productivity in the traded sector in a country by 1%, explains 0.36% increase in the long-run equilibrium real exchange rates, on an average, for that country. This is strong empirical support in favour of the Harrod-Balassa-Samuelson effect. The estimated terms of trade effect $(\beta_2)$ is also positive and significant, as expected.

As a final argument to support the HBS model, the panel is split into two parts- high income countries in one panel (which mostly include developed countries) and the middle- and low-income countries in a second panel (which mostly include emerging economies) and DOLS estimation results are presented in table.9.

The results show that the presence of HBS effect is robust to income/growth levels of countries. The HBS effect coefficients for high- and low-income countries are not statistically significantly different

from the overall panel coefficient in table 8 above. We can conclude that HBS effects are, on an average, prevalent in all types of countries, irrespective of their GDP growth rates or their relative income per capita status. The results show that the HBS model continues to be a powerful channel to explain the long-run dynamics of RER, not only for advanced economies, as is typically found in literature, but also for emerging market economies, for whom evidence remains weak in existing literature. TOT also shows up as a positive and significant long-run determinant for the real exchange rates in both types of countries.

**Table 9: Panel estimation results**

| | Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|---|
| High income countries | LRPRO | 0.370606** | 0.173608 | 2.13473 | 0.0345 |
| | LTOT | 0.806194*** | 0.287548 | 2.80368 | 0.0058 |
| Middle- and low-income countries | LRPRO | 0.347547*** | 0.131263 | 2.647713 | 0.0088 |
| | LTOT | 0.61239*** | 0.136326 | 4.492108 | 0.0000 |

**Notes: (1)** Dependent Variable: LREER **(2)** Method: Grouped Panel Dynamic Least Squares (DOLS) **(3)** *** denotes significant at 1%; **(4)** Cointegrating equation deterministics: constant, linear trend; **(5)** Fixed leads and lags specification (lead=1, lag=1) **(6)** Individual HAC standard errors & covariances.

# 7 CONCLUSION

This essay builds on the time honoured Harrod-Balassa-Samuelson effect, which explains long-run real exchange rate dynamics with traded sector productivity growth. Since 1964, many studies have used newer and evolving methods and tested this theory. Over the decades, a simple model presented by Balassa (1964) has been transformed, terms of trade and many other independent variables have been incorporated and econometric techniques such as panel cointegration for non-stationary dynamic panels have been developed and are widely used in current literature. This study tests the validity of the HBS theory in a new panel data sample at three levels- country-specific time-series, combined panel of all countries and separate panels of advanced and emerging economies. We employ currently prevalent time-series and panel data methods for cointegration and estimation. The results are strongly in favour of the HBS effects at each of the three levels tested.

The country-specific analysis was based on testing the HBS effect in a sample of 12 countries from across different income groups. The results showed that there is evidently a strong long-run equilibrium relationship between RER and the fundamentals in 11 out of 12 countries tested. The country-specific coefficient estimates of the HBS effect confirmed that productivity has positive and significant influence on RER in 10 out of those 12 countries.

Using panel-cointegration techniques for a mixed sample of 20 low-, middle- and high-income countries for the period 1991-2017, we find strong evidence for the HBS effects as one of the main driving forces underlying real exchange rate behaviour in the medium- to long-run. We obtain that

the positive relationship between the RER and relative productivity holds significantly in both advanced and emerging economies, with no almost similar average estimate of the HBS coefficient. The results also indicate that TOT constitutes a powerful determinant of the long-run dynamics of RER across countries.

It is also worth pointing out the main weaknesses of this analysis. First, the sample size in the country specific analysis is very small- only 27 years of data. Small sample size can also be a source of biasness in the estimated country-specific coefficients. Further, this implies that country specific results should be interpreted cautiously whereas panel-data results can be considered relatively more dependable due to a larger sample size. Length of time series has always been a major issue in macro-econometric testing for researchers since data availability, relevance and consistency of data varies considerably across countries. Second, only two independent variables have been used in explaining real exchange rates, which is perhaps not the best specification. Robustness of the results to inclusion of additional explanatory variables like net foreign assets or current account is not tested in this essay. Third, is the issue of the 'best' traded sector productivity measure. As described, value added per worker in agriculture and industry sectors has been used as the proxy in this essay. It is widely accepted that TFP is a better measure of productivity, however, sectoral TFP data is almost non-existent. Furthermore, agriculture and industry may not be the best proxies for the traded sector as many subsectors within these may very well be non-traded. Likewise, many sub-sectors in the services sector could be tradeable which are not considered in this analysis. This is especially true for most developing and low-income countries. However, sectoral data rarely exists for developing countries. Fourth, labour productivity as a productivity proxy could be influenced by permanent demand shocks. Finally, the theoretical model used in this essay is the most basic form of the HBS model, based on limiting assumptions like competitive markets, equal factor income shares for traded and non-traded sectors, perfect capital mobility (internationally) and complete factor price equalisation. While, relaxing some of these assumptions may make the analysis complex, the possibility of conclusions changing is not denied.

Last but not the least, in contemporary international macroeconomics literature monetary exchange rate theories and new open economy macroeconomic theories have become dominant. The once popular 'real' theories of exchange rates, one of which is the HBS theory, seem to have lost their importance in current literature. This essay does not contest the validity of monetary or NOEM theories, instead while acknowledging their role, it has attempted to show that the importance of real variables like productivity in explaining the failure of PPP and the dynamics long-run exchange rate cannot be undermined even today. Advances in econometric methods and improved accessibility to sector-level productivity data will only consolidate evidence in favour of the Harrod-Balassa-Samuelson theory.

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# APPENDICES

## APPENDIX 1: VARIABLE DESCRIPTION

The empirical model that is being tested in this essay is:

$$\ln \mathrm{RER_{it}} = \beta_0 + \beta_1 . \ln\left(\frac{\mathrm{A_{T,it}}}{\mathrm{A^*_{T,t}}}\right) + \beta_2 . \ln \mathrm{TOT_{it}} + \varepsilon_{\mathrm{it}}$$

This model is further expressed as:

$$\mathrm{LREER_{it}} = \beta_0 + \beta_1 . \mathrm{LRPRO_{it}} + \beta_2 . \mathrm{LTOT_{it}} + \varepsilon_{\mathrm{it}}$$

**LREER**

LREER is the is the natural log of real effective exchange rate (*REER*). *REER* is used as a proxy for real exchange rate (RER).

$$LREER_{it} = \ln REER_{it}$$

According to IMF, the CPI based real effective exchange rate (*REER*) of a country is the weighted geometric mean of its CPI relative to the CPI of its major trade partners.

$$REER_i = \prod_{i \neq j}^{n} \left[\frac{e_i P_i}{e_j P_j}\right]^{w_{ij}}$$

$n$ is the number of countries in the basket, in other words the number of trading partners of a country $i$. $j$ denotes the trade partners. $e_i$ is the country's exchange rate against the IMF's Special Drawing Rights (SDRs) in indexed form. $e_j$ is the exchange rate of trade partner country against the IMF's Special Drawing Rights (SDRs) in indexed form. $w_{ij}$is the geometric weight attached to the foreign currency *j*. $P_i$ is the consumer price index of the home country and $P_j$ is foreign trade partner's price index (CPI).

In this type of REER definition, an increase in the index denotes a real appreciation of the home currency, whereas a decrease implies a real depreciation.

Real effective exchange rate data is obtained from IMF-IFS via World Bank WDI database and uses 2010 as the base year (2010=100).

**LTOT**

LTOT is the natural log of the terms of trade (TOT) index. TOT is defined as the ratio of export prices to import prices.

$$LTOT_{it} = \ln TOT_{it}$$

$$TOT_i = \frac{P_{X,i}}{P_{M,i}}$$

$P_X$ is the export price and $P_M$ is the import price index for country $i$.

The data for TOT is also sourced from World Bank's WDI database from the series "Net Barter Terms of Trade." This series is compiled by World Bank from UNCTAD, Handbook of Statistics and data files, and IMF-IFS. TOT series is expressed an index of the ratio of export prices to import prices of a country with year 2000 as the base (2000 =100). Data for some countries was missing. The missing data was obtained from OECD National Accounts Statistics database on Terms of Trade and normalised with 2000 as the base year.

**LRPRO**

$\frac{Y_A}{L_A}$ = Labour Productivity in agriculture sector. The agriculture sector is defined as per ISIC Rev. 4 standards.

$\frac{Y_I}{L_I}$ = Labour Productivity in industry. The industry sector is defined as per ISIC Rev. 4 standards.

The traded sector is defined to include agriculture and industry. The non-traded sector then incudes the services sector and all other sectors in ISIC Rev. 4.

$\frac{Y_T}{L_T}$ = Labour Productivity in Traded Sector

In terms of the above definitions,

$$\frac{Y_T}{L_T} = \frac{Y_A}{L_A} + \frac{Y_I}{L_I}$$

$\left(\frac{Y_A}{L_A}\right)^W$ = World Average Labour Productivity in agriculture sector.

$\left(\frac{Y_I}{L_I}\right)^W$ = World Average Labour Productivity in industry.

$\left(\frac{Y_T}{L_T}\right)^W$ = World Average Labour Productivity in Traded Sector where

$$\left(\frac{Y_T}{L_T}\right)^W = \left(\frac{Y_A}{L_A}\right)^W + \left(\frac{Y_I}{L_I}\right)^W$$

$RPRO_i$ = Relative traded sector labour productivity in country $i$ i.e. traded sector labour productivity divided by world average traded sector labour productivity.

$$RPRO_i = \frac{\left(\frac{Y_T}{L_T}\right)^i}{\left(\frac{Y_T}{L_T}\right)^W} \times 100$$

*LRPRO* is the natural log of *RPRO*.

$$LRPRO_{it} = \ln RPRO_{it}$$

$LRPRO_{it}$ is the proxy for $\ln\left(\frac{A_{T,it}}{A^*_{T,t}}\right)$. In other words, traded sector labour productivity $\frac{Y_T}{L_T}$ is the proxy for $A_T$ and world average traded sector labour productivity $\left(\frac{Y_T}{L_T}\right)^W$ is the proxy for $A^*_T$.

Data for Labour productivity in agriculture $\frac{Y_A}{L_A}$ is obtained from World Bank WDI database from the series 'Value added per worker in agriculture forestry and fishing' This series is measured in constant 2010 US$ and compiled by World Bank from World Bank national accounts data and OECD National Accounts data files, and employment data from ILOSTAT.

Data for Labour productivity in industry $\frac{Y_I}{L_I}$ is obtained from World Bank WDI database from the series 'Value added per worker in industry (including construction).' This series is measured in constant 2010 US$ and compiled by World Bank from World Bank national accounts data and OECD National Accounts data files, and employment data from ILOSTAT.

Data for all variables is obtained and used under Creative Commons Attribution 4.0 International license (CC-BY 4.0).

## APPENDIX 2: SUMMARY OF ECONOMETRIC METHODS

### The Augmented Dickey Fuller Unit Root Test

Given a time-series $Y_t$, the ADF test checks if it has a unit root (i.e. it is non-stationary) by estimating the following model:

$$\Delta Y_t = \delta Y_{t-1} + \sum_{i=1}^{p} \gamma_i \Delta Y_{t-i} + u_t$$

And tests the following hypothesis.

$H_0: \delta = 0$ (unit root)

$H_1: \delta \neq 0$ (stationary)

The test statistic is the ADF τ-statistic, which is the usual t-statistic tested against ADF critical values. The optimal number of lags are selected on the basis of information criteria like Schwarz or Akaike to ensure that the error term $u_t$ is a pure white noise.

There are three specifications for deterministics in the above model (a) no drift no trend; (b) drift (constant) and (c) drift and linear trend.

### The two step Augmented Engle Granger Cointegration Test

The AEG test is a residual based cointegration test. Given non-stationary timeseries $y_t$ (dependent variable) and $\boldsymbol{x_t'}$ (independent variables), the first step to check if $y_t$ and $\boldsymbol{x_t'}$ are cointegrated is to estimate the following cointegrating equation using OLS and saving the residuals:

$$y_t = c + \boldsymbol{x_t'\beta} + u_t$$

The cointegrating equation can have the following deterministics (a) none; (b) constant; (c) linear trend; (d) quadratic trend. If the $u_t$ series is stationary, we can conclude that the time-series are cointegrated. Hence the second step is to perform an ADF type unit root test on the estimated residual series $\hat{u}_t$.

$$\Delta \hat{u}_t = \delta \hat{u}_{t-1} + \sum_{i=1}^{p} \gamma_i \Delta \hat{u}_{t-i} + v_t$$

The AEG test has the following null and alternative hypotheses.

$H_0: \delta = 0$ (no cointegration)

$H_1: \delta \neq 0$ (cointegration)

The test statistic is the Engle-Granger τ-statistic, which is the usual t-statistic tested against AEG critical values. The optimal number of lags $p$ are selected on the basis of information criteria like Schwarz or Akaike to ensure that the error term $v_t$ is a pure white noise.

### Stock and Watson's Dynamic Ordinary Least Squares Estimation

Dynamic OLS (DOLS) method involves augmenting the cointegrating regression with lags and leads of the independent variables so that the resulting cointegrating equation error term is orthogonal to the entire history of the stochastic regressor innovations. According to Stock and Watson (1993), if $I(1)$ time series are cointegrated, then OLS is super consistent but is non-normally distributed, hence usual hypotheses testing cannot be done. The DOLS estimator is consistent, more efficient than OLS, when used with HAC standard errors and the t-statistics have a standard normal distribution asymptotically.

If $y_t$ *and* $x_t$ are cointegrated non-stationary time-series, then the DOLS estimation method is simply estimating the following model using OLS:

$$y_t = c + \beta_1 x_t + \sum_{i=-p}^{+q} \delta_i \Delta x_{t+i} + u_t$$

Here p are the lags and q are the leads of the first diff. of the independent variable. The lags and leads can be selected on the basis on information criteria like Schwarz or Akaike. Deterministics can include (a) none; (b) constant; (c) linear trend; (d) quadratic trend.

**Auto Regressive Distributed Lag (ARDL) Model and Bounds Cointegration Test (Pesaran, Shin and Smith, 2001)**

This model was developed by Pesaran et al. (2001) and can be applied to test for cointegration between time series irrespective of the order of their integration (irrespective of whether regressors are purely $I(0)$, or $I(1)$or mutually co-integrated). This approach is also useful in small sample estimation. The ARDL-OLS estimator of the long-run cointegrating vector is super consistent and asymptotically normally distributed irrespective of the order of integration of the regressors. The ARDL modeling approach involves estimating the following conditional error correction model:

$$\Delta y_t = c + \sum_{i=1}^{p-1} \gamma_i \, \Delta y_{t-i} + \sum_{j=0}^{q-1} \psi_{1j} \, \Delta x_{1,t-j} + \sum_{j=0}^{q-1} \psi_{2j} \, \Delta x_{2,t-j} + \theta_0 y_{t-1} + \theta_1 x_{1,t-1} + \theta_2 x_{2,t-1} + u_t$$

Here p and q are the optimal number of lags in the dependent and independent variables, in the underlying levels ARDL model. The optimal $ARDL(p, q, q)$ model can be selected using information criteria like Akaike or Schwarz.

Once the model is estimated using OLS, the bounds test involves two hypotheses tests. The first one has the following null and alternative hypotheses:

$H_0: \theta_0 = \theta_1 = \theta_2 = 0$ (No long-run levels relationship)

$H_1: \theta_0 \neq \theta_1 \neq \theta_2 \neq 0$ (Long-run levels relationship)

As in conventional cointegration testing, here we are testing for the absence of a long-run equilibrium relationship between the variables. This absence coincides with zero coefficients for $y_{t-1}, x_{1,t-1}$ and $x_{2,t-1}$ in the above equation. A rejection of $H_0$ implies that we have a long-run relationship.

We have a non-standard "F-test" for this purpose with the usual F-Statistic. Exact critical values for the F-test aren't available for an arbitrary mix of $I(0)$ and $I(1)$variables. However, Pesaran *et al.* (2001) supply bounds on the critical values for the asymptotic distribution of the F-statistic. In each case, the lower bound is based on the assumption that all of the variables are $I(0)$, and the upper bound is based on the assumption that all of the variables are $I(1)$.

If the computed F-statistic falls below the lower bound we would conclude that the variables are $I(0)$, so no cointegration is possible, by definition. If the F-statistic exceeds the upper bound, we conclude that we have cointegration. Finally, if the F-statistic falls between the bounds, the test is inconclusive.

Pesaran *et al.* (2001) supply bounds for the asymptotic distribution of the F-statistic. Kripfganz and Schneider (2018) have provided critical value bounds for finite samples along with approximate p-value bounds which are used in this essay.

A second test is the bounds t-test with

$H_0: \theta_0 = 0$ (No long-run levels relationship)

$H_1: \theta_0 < 0$ (Long-run levels relationship)

If the t-statistic for the coefficient of $y_{t-1}$ is greater than the $I(1)$bound tabulated by Pesaran *et al.* (2001), this would support the conclusion that there is a long-run relationship between the variables. If the t-statistic is less than the $I(0)$ bound, we conclude that the data are all stationary. Kripfganz and Schneider (2018) have provided critical value bounds for finite samples along with approximate p-value bounds which are used in this essay. Note that $\theta_0$ denotes the adjustment parameter i.e. it is that rate of adjustment towards long-run equilibrium, in case the variables are is out of equilibrium.

In case both the F-test and the t-test reject no cointegration, we can proceed with estimating the long-run cointegrating vector using ARDL-OLS approach. This involves estimating the ARDL model in levels.

$$y_t = c + \sum_{i=1}^{p} \phi_i \, y_{t-i} + \sum_{j=0}^{q} \beta_{1j} \, x_{1,t-j} + \sum_{j=0}^{q} \beta_{2j} \, x_{2,t-j} + u_t$$

From this estimation, the speed of adjustment $\alpha$ is given by $\alpha = 1 - \sum_{i=1}^{p} \phi_i$. Note that $\alpha = \theta_0$ from the previous equation.

The long-run coefficients of $x_{1t}$ and $x_{2t}$ are respectively given as $\beta_1 = \frac{\sum_{j=1}^{q} \beta_{1j}}{\alpha}$ and $\beta_2 = \frac{\sum_{j=1}^{q} \beta_{2j}}{\alpha}$.

The optimal lags are selected on the basis of information criteria like Akaike or Schwarz. The ARDL model has five deterministic specifications: (1) no constant, no trend; (2) restricted constant, (3) constant; (4) restricted trend; (5) trend. Table 2 in section 7.1.3 shows which specification is used for which country.

**Im, Pesaran and Shin Panel Unit Root Test (2003)**

IPS Panel unit root test employs an ADF type approach to the panel setting. It involves performing separate ADF regressions for each cross-section:

$$\Delta Y_{it} = \delta_i Y_{i,t-1} + \sum_{j=1}^{p_i} \gamma_{ij} \Delta Y_{i,t-j} + u_{it}$$

And tests the following hypothesis.

$H_0: \delta_i = 0 \, , \forall \, i$

$H_1: \delta_i \neq 0$ , for at least one $i$

The test statistic is the IPS W-stat obtained after averaging and normalising the t-statistics for all cross-sections. It is tested against IPS critical values. The optimal number of lags are selected on the basis of information criteria like Schwarz or Akaike to ensure that the error term $u_{it}$ is a pure white noise. There are three deterministics specifications for the above model (a) no drift no trend; (b) individual drift (constant) and (c) individual drift and linear trend. See Baltagi (2005) for more.

**Breitung Test (2000)**

Breitung test applies a common unit root approach. The first step in the Breitung test is estimation of ADF type regressions as in IPS test above but the maintained hypothesis imposes a common root $\delta$:

$$\Delta Y_{it} = \delta Y_{i,t-1} + \sum_{j=1}^{p_i} \gamma_{ij} \Delta Y_{i,t-j} + u_{it}$$

Breitung test statistic is obtained after several normalisations. See Baltagi (2005) for more details. The hypotheses in this test are:

$H_0: \delta = 0$ (Panel has a unit root)

$H_1: \delta \neq 0$ (No unit root in panel)

**Kao Cointegration test (1999)**

Kao test is a residual based ADF/AEG type test. Given non-stationary panel series $y_{it}$ (dependent variable) and $\boldsymbol{x'_{it}}$ (independent variables), the first step to check if $y_{it}$ and $\boldsymbol{x'_{it}}$ are cointegrated is to estimate the following cointegrating equation using OLS and saving the residuals:

$$y_{it} = \alpha_i + \boldsymbol{x'_{it}\beta} + u_{it}$$

The Cointegrating equation can have the following deterministics (a) none; (b) constant; (c) linear trend; If the $u_{it}$ series is stationary, we can conclude that the variables are panel cointegrated. Hence the second step is to perform an ADF type unit root test on the estimated residual series $\hat{u}_{it}$.

$$\hat{u}_{it} = \rho \hat{u}_{i,t-1} + \sum_{j=1}^{p} \gamma_j \Delta \hat{u}_{i,t-j} + v_{it}$$

The Kao test has the following null and alternative hypotheses.

$H_0: \rho = 0$ (no cointegration)

$H_1: \rho \neq 0$ (cointegration)

The test statistic is the Kao t-statistic, which is the usual t-statistic tested against Kao t-critical values. The optimal number of lags $p$ are selected based on information criteria like Schwarz or Akaike to ensure that the error term $v_{it}$ is a pure white noise.

**Pedroni Tests (2004)**

Pedroni (2004) proposes a residual-based test for the null of no cointegration for panels with multiple regressors in which the short-run dynamics and the long-run slope coefficients are permitted to be heterogeneous across individuals. The test allows for individual heterogeneous fixed effects and trend terms and no exogeneity requirements are imposed on the regressors of the cointegrating regressions.

The first step involves estimating the cointegrating equation for each cross-section.

$$y_{it} = \alpha_i + \boldsymbol{x'_{it}\beta_i} + u_{it}$$

Note that the cointegrating vector $\beta$ is allowed to vary across cross-sections so that considerable heterogeneity is allowed in this model specification.

Pedroni considers seven residual-based panel cointegration statistics, four based on pooling the data within-dimension (panel cointegration statistics) and three based on pooling between-dimension (group mean cointegration statistics).

The within-dimension tests and between dimension tests have different alternative hypotheses. The set of null and alternative hypotheses is given as:

$H_0: \rho_i = 1 \; \forall \; i$

$H_1^w: \rho_i = \rho < 1 \;, \forall \; i$ (within dimension alternative hypothesis)

$H_1^b: \rho_i < 1 \;, \forall \; i$ (between dimension alternative hypothesis)

The panel cointegration statistics impose a common AR coefficient under the alternative hypothesis. The group mean cointegration statistics allow for heterogeneous AR coefficients under the alternative hypothesis. The tests based on the between dimension are more general as they allow for cross-section heterogeneity.

The details of the 7 statistics are omitted here. See Baltagi (2005) for more details. In this essay, the parametric ADF statistic and the non-parametric PP statistic are reported for both the within and between dimension cases.

**Panel Dynamic Ordinary Least Squares**

Kao and Chiang (2000) and Pedroni (2001) have extended the DOLS estimator to the panel setting. This essay uses Pedroni group mean PDOLS estimation technique for panel cointegrated series. The PDOLS estimator is estimated using OLS from:

$$y_{it} = \alpha_i + \beta_i x_{it} + \sum_{j=-p_i}^{+q_i} \delta_{ij} \Delta x_{i,t+j} + u_{it}$$

Here $p_i$ are the lags and $q_i$ are the leads of the first diff. of the independent variable. The lags and leads can be selected on the basis on information criteria like Schwarz or Akaike. Deterministics can include (a) none; (b) constant; (c) linear trend; (d) quadratic trend.

The homogeneous long-run cointegrating estimate is obtained by averaging over cross-sectional estimates of $\beta_i$'s. The procedure thus allows for heterogeneous cointegrating vectors, short run dynamics and deterministics (constant or trend) across cross-sections.